Linux Commands

1. kill – Terminate a process by PID

Command:

```
sleep 300 &
jobs -l
kill 819
```

Explanation: Starts a test sleep process in the background, lists its job and PID, then sends a termination signal to PID 819.

Observed result: Your screenshot showed PID 819 running, then the shell reported the sleep process as Terminated. Always verify a PID before using kill.

2. ip link – Display network interfaces

Command:

```
ip link
```

Explanation: Displays network interfaces, their MAC addresses, MTU, and link state.

Observed result: Your WSL2 system showed lo (loopback) and eth0. eth0 was UP.

3. getent hosts – Resolve a hostname

Command:

```
getent hosts google.com
```

Explanation: Looks up host information using configured name-resolution sources, including DNS.

Observed result: The output returned an IPv6 address for google.com: 2404:6800:4007:8f::200e. DNS answers can vary.

4. head – Display the first lines

Command:

```
head -n 5 file1.txt
```

Explanation: Displays the first five lines of file1.txt. -n specifies the number of lines.

Observed result: The screenshot showed Hi, Hello, How, are, and the line beginning with you?. The file is not modified.

5. wc – Count lines, words, and bytes

Command:

```
wc file1.txt
```

Explanation: Displays line count, word count, and byte count, followed by the filename.

Observed result: Output: 5 11 46 file1.txt — 5 lines, 11 words, and 46 bytes.

6. logger – Send a message to the system log

Command:

logger "Linux practice test from Srinivas"

Explanation: Sends a custom message to the system logging service. Useful for testing application logging.

Observed result: Your custom message appeared in journal output as “Linux practice test from Srinivas”.

7. journalctl – View recent system logs

Command:

journalctl -n 5

Explanation: Displays the five most recent journal entries.

Observed result: Your output included service messages, a WSL warning, and your logger message. The shell showed the pager as Stopped, likely because Ctrl+Z was pressed. Use q to exit the pager; use fg to resume a suspended pager.

8. chown – Change owner and group

Command:

sudo chown root:root file1.txt
ls -l file1.txt

Explanation: Changes both the owner and group of file1.txt to root. ls -l verifies the result.

Observed result: The listing showed owner root and group root.

9. chown – Restore current user ownership

Command:

sudo chown "$USER":"$(id -gn)" file1.txt
ls -l file1.txt

Explanation: Restores the owner to the current user and the group to the user’s primary group. $USER gives the username; id -gn gives the primary group.

Observed result: The final listing showed owner Srini and group Srini, confirming restoration.

10. dmesg – Display kernel messages

Command:

sudo dmesg -T | tail -15

Explanation: dmesg displays kernel messages; -T makes timestamps readable. The pipe sends output to tail, which shows the last 15 lines.

Observed result: Your output included WSL2/Hyper-V and graphics-related messages. Environment-specific warnings do not automatically mean the system has failed.

11. uptime – Display running time

Command:

uptime -p

Explanation: Displays system uptime in a human-readable format. -p means pretty format.

Observed result: Output: up 2 minutes. Uptime changes as the system continues running.

12. pstree – Display processes in a tree

Command:

pstree -p

Explanation: Displays processes in a parent-child tree. -p includes process IDs.

Observed result: Your output showed systemd and child processes such as cron, dbus-daemon, rsyslogd, systemd-journald, and the terminal shell.

13. pidstat – Monitor process statistics

Command:

pidstat 1 3

Explanation: Displays process statistics every one second for three reports. It reports process IDs and CPU usage information.

Observed result: Your output showed packagekitd with PID 766. The first sample showed about 1% CPU; the displayed average was about 0.33%. Values depend on system activity.